Program 5 - Implement email service using NodeJS nodemailer service.

Below is a **complete lab-manual style, step-by-step guide** to **implement an Email Service using Node.js with Nodemailer**, written in the **same format and depth** as your previous programs, and **integrated into the same project folder**.

This is **exam-oriented, beginner-friendly**, and tested for **Windows 11 + VS Code**.

Implementing Email Service using Node.js (Nodemailer)

Platform: Windows 11
Editor: Visual Studio Code
Language: JavaScript (Node.js)

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1. Project Folder Structure (Integrated)

Continue using the **same folder** you already created.

Folder Name

node-basics

Folder Structure

node-basics

```
│
├── server.js
├── fileIO.js
├── queryString.js
├── routing.js
├── email.js        (NEW – Email service)
├── package.json
```

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2. Prerequisite: Enable Email Access

Option Used (Recommended for Lab)

✓ Gmail with App Password

⚠️ Normal Gmail password will NOT work

Step 1: Enable 2-Step Verification (One-time)

1. Go to **Google Account → Security**
2. Enable **2-Step Verification – Turn ON**

Step 2: Generate App Password

1. Google Account → Security
2. Search **App Passwords**
3. Select:
    - App: **Mail**
    - Device: **Windows Computer**
4. Copy the **16-character password**

📌 Keep it safe — you will paste it in code.

How to Find "App Passwords" in Google Account (Updated – 2025)

⚠ **Important prerequisites**

- You **MUST** have:
    - A **Gmail account**
    - **2-Step Verification enabled**
- App Passwords **will NOT appear** otherwise

STEP 1: Open Google Account Security Page

Open this link in browser (recommended):

👉 **https://myaccount.google.com/security**

Log in with your Gmail account.

STEP 2: Enable 2-Step Verification (Mandatory)

Scroll to **"How you sign in to Google"**

1. Click **2-Step Verification**
2. Follow instructions (OTP / Authenticator / SMS)
3. Make sure it shows:
4. Status: ON

📌 **Without this, App Passwords will NEVER appear**

STEP 3: Find App Passwords (Hidden Location)

After 2-Step Verification is ON:

Method 1 (Most Reliable)

1. Go to **Security**
2. At the top **search box**, type:

App passwords

✓ You should see **App passwords** in search results
Click it.

Method 2 (Manual Navigation)

1. Security
2. How you sign in to Google
3. Click **2-Step Verification**
4. Scroll to the **bottom**
5. You'll see:

App passwords

👉 Click it

STEP 4: Generate App Password

Once inside **App Passwords**:

1. Select App:
2. Mail

3. Select Device:
4. Windows Computer
5. Click **Generate**

STEP 5: Copy the Password

You'll see something like:

abcd efgh ijkl mnop

✓ This is the **16-character App Password**
✓ Copy it **without spaces**

Example:

abcdefghijklmnop

📌 This replaces your Gmail password in Node.js code.

STEP 6: Use It in Your Code

In email.js:

```
auth: {
   user: 'your_email@gmail.com',
   pass: 'abcdefghijklmnop'
}
```

❌ DO NOT use your normal Gmail password
❌ DO NOT include spaces

🚨 **If You STILL Do NOT See "App Passwords"**

This happens in these cases:

❌ **Google Workspace / College Email**

If your email looks like:

name@college.edu

👉 App Passwords are **disabled by admin**

☑ **Solution**: Use a **personal Gmail account**

❌ **Recently Created Google Account**

Google sometimes **blocks App Passwords temporarily**

☑ Wait **24–48 hours** after enabling 2-Step Verification

❌ **Logged in via Child / Family Account**

App Passwords are **not allowed**

🔁 **Alternative (If App Password is Impossible)**

If App Passwords are **not available**, I can give you:

Option 1 (Recommended for Lab)

✓ **Ethereal Email (Fake SMTP – no real email needed)**
✓ Works instantly
✓ Ideal for exams & lab records

Option 2

✓ Outlook / Microsoft email SMTP

👉 Tell me **"Give alternative email method"** and I'll provide it.

☑ **Quick Checklist**

Check	Status
2-Step Verification ON	✓
Personal Gmail	✓
Search "App passwords"	✓
Generated 16-digit password	✓

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3. Initialize Node Project (IMPORTANT)

Open **VS Code Terminal** inside node-basics folder.

npm init -y

✓ This creates package.json

This creates the error

this creates the error, npm : File C:\Program Files\nodejs\npm.ps1 cannot be loaded because running scripts is disabled on this system. For more information, see about_Execution_Policies at https:/go.microsoft.com/fwlink/?LinkID=135170. At line:1 char:1 + npm init -y + ~~~ + CategoryInfo : SecurityError: (:) [], PSSecurityException + FullyQualifiedErrorId : UnauthorizedAccess

Solution - This is a **very common Windows + PowerShell issue**, and you've done **nothing wrong** 👍
The error is **NOT related to Node.js or npm** — it's caused by **Windows PowerShell security policy**.

I'll explain **why it happens**, then give you **two safe, step-by-step solutions**.
👉 Follow **Solution 1** (recommended for labs).

❌ **Why this error occurs**

Error message:

npm.ps1 cannot be loaded because running scripts is disabled on this system

Reason:

- VS Code terminal is using **PowerShell**
- PowerShell **blocks script execution** by default
- npm internally runs a script: npm.ps1
- PowerShell blocks it → ❌ error

📌 This is a **Windows security feature**, not an npm problem.

☑ **SOLUTION 1 (RECOMMENDED): Change Execution Policy (SAFE)**

This allows npm to run **only for your user account**.

Step 1: Open PowerShell as Administrator

1. Click **Start**
2. Type **PowerShell**
3. Right-click **Windows PowerShell**
4. Click **Run as Administrator**

Step 2: Check Current Policy (Optional)

Get-ExecutionPolicy

Likely output:

Restricted

Step 3: Set Execution Policy (SAFE OPTION)

Run this command:

Set-ExecutionPolicy RemoteSigned -Scope CurrentUser

Step 4: Confirm

When prompted:

Do you want to change the execution policy?

Type:

Y

Press **Enter**

Step 5: Close PowerShell

Close the Administrator PowerShell window.

Step 6: Restart VS Code

✓ Close VS Code completely
✓ Open VS Code again
✓ Open node-basics folder

Step 7: Run npm command again

Open VS Code Terminal and run:

npm init -y

☑ **Expected Output:**

Wrote to package.json

🎉 Problem solved permanently.

☑ **SOLUTION 2 (TEMPORARY / QUICK FIX)**

If you **do not want to change security policy**:

Step 1: Change Terminal to Command Prompt

In VS Code:

1. Click **Terminal**
2. Click **New Terminal**
3. Click the dropdown ▼
4. Select **Command Prompt (cmd)**

Step 2: Run npm command

npm init -y

✓ This works because **cmd does not block scripts**

⚠ But PowerShell will still fail later.

🔍 **Why Solution 1 is Better (For Exams & Labs)**

Reason	Explanation
Permanent fix	Works for all Node projects
Safe	Applies only to current user
Industry standard	Used by developers
Avoids confusion	PowerShell works normally

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4. Install Nodemailer Package

Run:

npm install nodemailer

✓ This installs Nodemailer in node_modules

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5. Create Email Program File

Step 1: Create File

1. VS Code → Explorer
2. Click **New File**
3. Name it:

email.js

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6. Write Email Sending Program

email.js

```
// Import nodemailer
const nodemailer = require('nodemailer');


// Create transporter object
const transporter = nodemailer.createTransport({
   service: 'gmail',
```

```
  auth: {
    user: 'your_email@gmail.com',       // sender email
    pass: 'your_app_password'           // app password
  }
});

// Email options
const mailOptions = {
  from: 'your_email@gmail.com',
  to: 'receiver_email@gmail.com',
  subject: 'Node.js Email Test',
  text: 'This email is sent using Node.js Nodemailer service.'
};

// Send email
transporter.sendMail(mailOptions, (error, info) => {
  if (error) {
    console.log("Error:", error);
  } else {
    console.log("Email sent successfully!");
    console.log("Response:", info.response);
  }
});
```

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7. Replace Email Credentials (VERY IMPORTANT)

Replace:

your_email@gmail.com

With:

👉 Your Gmail address

Replace:

your_app_password [**motd vnca ownc imxb**] without space

Replace:

to: 'receiver_email@gmail.com',

with:

receivermail@gmail.com

With:

👉 16-digit **App Password**

⚠️ **Do NOT use your Gmail login password**

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8. Run the Email Program

Step 1: Open Terminal

Ctrl + `

Step 2: Execute Program

node email.js

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9. Expected Output

Terminal Output

Email sent successfully!

Response: 250 2.0.0 OK

Inbox Output

📧 Receiver will receive the email instantly.

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10. Program Explanation (Exam-Ready)

Statement	Explanation
require('nodemailer')	Imports nodemailer module
createTransport()	Creates email transport service
service: 'gmail'	Uses Gmail SMTP
auth	Email authentication
mailOptions	Email details
sendMail()	Sends the email

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11. Common Errors & Solutions

❌ **Invalid login**

✓ Use **App Password**, not Gmail password
✓ Ensure 2-Step Verification enabled

❌ **Module not found**

Cannot find module 'nodemailer'

✓ Run:

npm install nodemailer

❌ **Less secure app error**

✓ App Password solves this

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12. Viva / Lab Questions

1. What is Nodemailer?
2. Why App Password is required?
3. What is SMTP?
4. Purpose of createTransport()?
5. Difference between text and html email?

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13. Lab Record Format (Optional)

Aim:
To implement email service using Node.js Nodemailer.

Algorithm:

1. Install Nodemailer
2. Create transporter
3. Configure mail options
4. Send email

Result:
Email sent successfully using Node.js.

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14. Optional Enhancements

✓ HTML email
✓ Attachments
✓ Email using Outlook
✓ Integrate with HTTP server (send email via browser)

motd vnca ownc imxb